

Convex Optimization - Project 1

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$$a) L(p) = \sum_{\{i,j\} \in S} d(x_i, x_j) + \sum_{\substack{\{i,j\} \\ \in D}} \frac{1}{d(x_i, x_j)}$$

$$d(x_i, x_j) = (x_i - x_j)^T p (x_i - x_j), \quad p \in S_+^n$$

↳ affine on p

$L(p)$ is convex because it consists of two terms in which

First term is sum of affine functions

and second term is sum of convex functions

(because invpos of nonnegative affine function is convex
p.p.)

So the whole term is convex)

and then sum of two terms will be convex $\rightarrow p$ such that
we will find it minimizes $L(p)$.

b, c) The code for part B and C is attached.